

Consider your group is assigned to a project and this group needs to maintain the codebase using Git and GitHub by doing following tasks –

1. Repository Initialization

Problem: You're starting a new project. How do you set up Git to track changes in your project directory?
Add two files in it and perform following -

- create a new branch dev
- modify a file & commit on the dev branch
- merge dev into main
- push the merged main to the remote repository
- clone a remote repository to the folder "myCode".

```
kaver@KAVERI MINGW64 ~
$ cd D:

kaver@KAVERI MINGW64 /d
$ cd MCA/"Sem II"/"DevOps Lab"

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Pract2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Pract2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Pract2/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ echo "This is file 1" > file1.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ echo "This is file 2" > file2.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git add .
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git commit -m "Initial commit"
[master (root-commit) 9e7799a] Initial commit
2 files changed, 2 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git branch dev

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git checkout dev
Switched to branch 'dev'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (dev)
$ echo "File 1 updated!" >> file1.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (dev)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (dev)
$ git commit -m "Modified file1 on dev branch"
[dev a5a6723] Modified file1 on dev branch
1 file changed, 1 insertion(+)
```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (dev)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git merge dev
Updating 9e7799a..a5a6723
Fast-forward
 file1.txt | 1 +
 1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git remote add origin https://github.com/kaverikadamm/pract2.git

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git push -u origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 550 bytes | 110.00 kib/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/pract2.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Pract2 (master)
$ git clone https://github.com/kaverikadamm/pract2.git myCode
Cloning into 'myCode'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 7 (delta 0), reused 7 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.

```

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * kaverikadamm / Repository name * pract2
pract2 is available.

Great repository names are short and memorable. How about [shiny-comscoms?](#)

Description
0 / 350 characters

2 Configuration

Choose visibility *
Choose who can see and commit to this repository Private

Add README
READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

2. Configure and Commit

Problem: Create a repository, configure username and email locally for the repo, create a commit, then show the commit log.

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir q2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd q2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/q2/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ git config user.name "Kaverikadam"

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ git config user.email "a250007@famt.ac.in"

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ echo "This is sample file" >> sample.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ git add sample.txt
warning: in the working copy of 'sample.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ git commit -m "First Commit"
[master (root-commit) 2f4351a] First Commit
1 file changed, 1 insertion(+)
create mode 100644 sample.txt

```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (master)
$ git log
commit 2f4351a725ac9cfba39f7645f1da1f71e1ca9965 (HEAD -> master)
Author: Kaverikadam <a250007@famt.ac.in>
Date: Mon Mar 23 22:14:24 2026 +0530

    First Commit

```

3. Recover a Deleted Branch

Problem: A teammate deleted the `feature/payment` branch, but you need to restore it. How do you recover from it?

```

$ git checkout -b feature/payment
Switched to a new branch 'feature/payment'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (feature/payment)
$ echo "Payment module implementation" >> test.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (feature/payment)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (feature/payment)
$ git commit -m "Added payment module"
[feature/payment b9e4ef7] Added payment module
1 file changed, 1 insertion(+)
create mode 100644 test.txt

```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (feature/payment)
$ git remote add origin https://github.com/kaverikadamm/pract2.git

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/q2 (feature/payment)
$ git push -u origin feature/payment
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 509 bytes | 509.00 kib/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/payment' on Github by visiting:
remote: https://github.com/kaverikadamm/pract2/pull/new/feature/payment
remote:
To https://github.com/kaverikadamm/pract2.git
 * [new branch] feature/payment -> feature/payment
branch 'feature/payment' set up to track 'origin/feature/payment'.

```



```
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (master)
$ git branch -D feature/payment
Deleted branch feature/payment (was b9e4ef7).

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (master)
$ git reflog
2f4351a (HEAD -> master) HEAD@{0}: checkout: moving from feature/payment to master
b9e4ef7 (origin/feature/payment) HEAD@{1}: commit: Added payment module
2f4351a (HEAD -> master) HEAD@{2}: checkout: moving from master to feature/payment
2f4351a (HEAD -> master) HEAD@{3}: commit (initial): First Commit
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (master)
$ git checkout -b feature/payment b9e4ef7
Switched to a new branch 'feature/payment'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (feature/payment)
$ git branch
* feature/payment
  master

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (feature/payment)
$ ls
sample.txt  test.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q2 (feature/payment)
$ cat test.txt
Payment module implementation
```

4. Fix a Broken Merge

Problem:

1. Merging `feature/login` into `main` caused conflicts in `login.py`. How do you resolve them?
2. Describe steps to resolve a merge conflict when two branches modify the same line in an HTML file. Include Git commands to: (a) identify conflicting files, (b) edit the file to resolve, (c) finalize the merge.

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q4

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q4

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q4/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ echo "print('Login v1')" > login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git add login.py
warning: in the working copy of 'login.py', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git commit -m "Initial commit with login.py"
[master (root-commit) 8ec25b5] initial commit with login.py
1 file changed, 1 insertion(+)
create mode 100644 login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git checkout -b feature/login
Switched to a new branch 'feature/login'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (feature/login)
$ echo "print('Login feature added')" >> login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (feature/login)
$ git add login.py
warning: in the working copy of 'login.py', LF will be replaced by
CRLF the next time Git touches it
```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (feature/login)
$ git commit -m "Updated login.py in feature branch"
[feature/login 64e8d04] Updated login.py in feature branch
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (feature/login)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ echo "print('Login updated in master')" >> login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git add login.py
warning: in the working copy of 'login.py', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git commit -m "Updated login.py in master branch"
[master f42f042] Updated login.py in master branch
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git merge feature/login
Auto-merging login.py
CONFLICT (content): Merge conflict in login.py
Automatic merge failed; fix conflicts and then commit the result.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   login.py

no changes added to commit (use "git add" and/or "git commit -a")

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master|MERGING)
$ cat login.py
print('Login v1')
<<<<<<< HEAD
print('Login updated in master')
=====
print('Login feature added')
>>>>>> feature/login

```

Open login.py and modify as:

```

login.py  X
D: > MCA > SEM II > DevOps Lab > Q4 > login.py
1  print('Login updated in master + feature added')

```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master|MERGING)
$ git add login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master|MERGING)
$ git commit -m "Resolved Merge conflict in login.py"
[master 62cd8b6] Resolved Merge conflict in login.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q4 (master)
$ git status
On branch master
nothing to commit, working tree clean

```

2)

```
$ cd Demo

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Demo/.git/

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ echo "<h1>Main Page</h1>" > index.html

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced
by CRLF the next time Git touches it
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git commit -m "Initial HTML"
[master (root-commit) 0f15953] Initial HTML
 1 file changed, 1 insertion(+)
 create mode 100644 index.html

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git checkout -b feature
Switched to a new branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (feature)
$ echo "<h1>Feature Page</h1>" > index.html

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (feature)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced
by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (feature)
$ git commit -m "Feature HTML change"
[feature 15bfe1c] Feature HTML change
 1 file changed, 1 insertion(+), 1 deletion(-)

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (feature)
$ git checkout master
Switched to branch 'master'
```

```
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ echo "<h1>Main Updated</h1>" > index.html

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced
by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git commit -m "Main HTML change"
[master c048a09] Main HTML change
1 file changed, 1 insertion(+), 1 deletion(-)

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master)
$ git merge feature
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit the result.

$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

```
<<<<<< HEAD
<h1>Main Updated</h1>
-----
<h1>Feature Page</h1>
>>>>>> feature
```

```
HEAD
<h1>Main Updated</h1>
<h1>Feature Page</h1>
feature
```

```
kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master|MERGING)
$ git add index.html

kaver@KAVERI MINGW64 /d/MCA/SEM II/DevOps Lab/Demo (master|MERGING)
$ git commit -m "Resolved HTML merge conflict"
[master 9cd6f44] Resolved HTML merge conflict
```

5. Track Down a Bug's Origin

Problem: Line 42 in `utils.js` is causing a crash. Who wrote that line, and in which commit?

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ touch util.js
```




```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git add util.js
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git commit -m "A buggy commit"
[master 145b46b] A buggy commit
1 file changed, 43 insertions(+)
create mode 100644 util.js
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git blame util.js
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 1) 1)
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 2) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 3) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 4) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 5) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 6) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 7) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 8) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 9) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 10) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 11) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 12) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 13) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 14) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 15) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 16) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 17) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 18) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 19) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 20) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 21) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 22) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 23) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 24) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 25) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 26) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 27) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 28) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 29) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 30) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 31) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 32) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 33) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 34) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 35) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 36) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 37) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 38) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 39) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 40) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 41) .
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 42) write bug here undefined function()
145b46b7 (KaveriKadam 2026-03-26 22:34:19 +0530 43)
```



```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git show 145b46b7
commit 145b46b7490bf07fc3b177934f61d87f91bcefb2 (HEAD -> master)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 22:34:19 2026 +0530

    A buggy commit

diff --git a/util.js b/util.js
new file mode 100644
index 0000000..f9d3783
--- /dev/null
+++ b/util.js
@@ -0,0 +1,43 @@
+1)
+.
+.
+.
+.
+.
+.

```

6. Dealing with a Remote Repository

Problem: Write git commands to: create local repo project-alpha, add notes.txt ("Meeting Agenda") and empty data.csv, commit, create branch experiment, add test.py with print("Hello"), commit. Push the entire repo to the remote repository and then modify notes.txt.

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir project-alpha

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd project-alpha

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/project-alpha/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ echo "Meeting Agenda" >> notes.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ touch data.csv

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ git add .
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ git commit -m "Added notes.txt and empty data.csv"
[master (root-commit) eb8e476] Added notes.txt and empty data.csv
2 files changed, 1 insertion(+)
create mode 100644 data.csv
create mode 100644 notes.txt

```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ git checkout -b experiment
Switched to a new branch 'experiment'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ echo "print('Hello')>>test.py
> /
> echo "print('Hello')>>test.py
bash: syntax error near unexpected token `('

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ echo "print('Hello')>>test.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git add test.py
warning: in the working copy of 'test.py', LF will be replaced by CRLF the next time git touches it
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git commit -m "Added test.py in experiment branch"
[experiment 573ad31] Added test.py in experiment branch
1 file changed, 1 insertion(+)
create mode 100644 test.py
```

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * / Repository name *

project-alpha is available.

Great repository names are short and memorable. How about [automatic-octo-fishstick](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility *

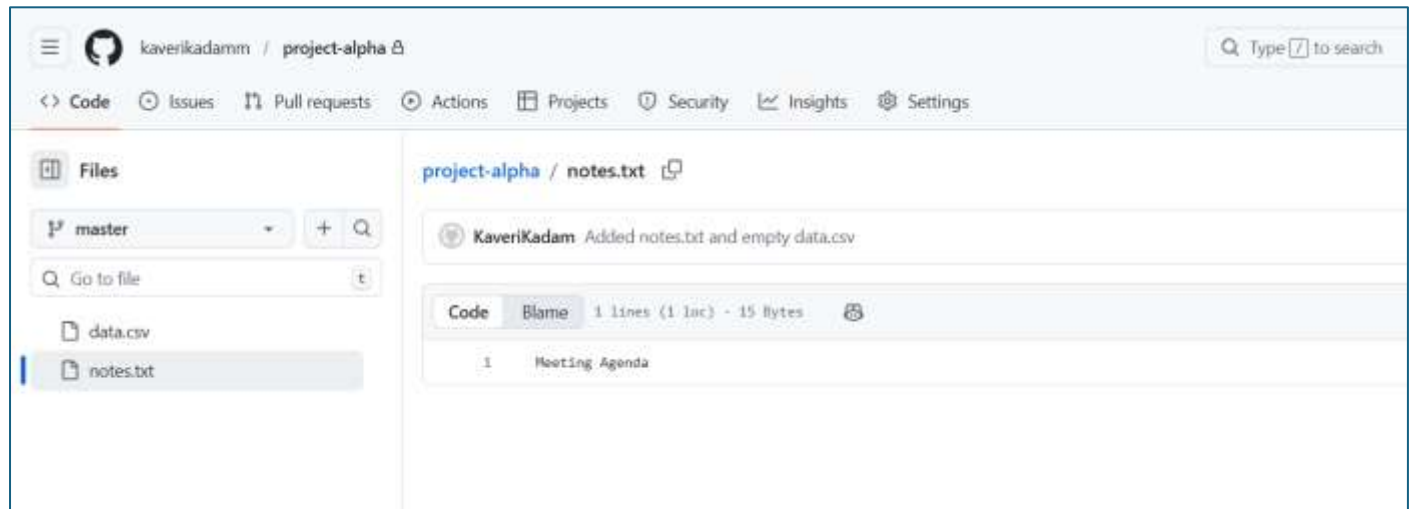
Choose who can see and commit to this repository

☒ Private

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git remote add origin https://github.com/kaverikadamm/project-alpha.git

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git push -u origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 284 bytes | 56.00 kib/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/project-alpha.git
* [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git push origin experiment
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 339 bytes | 339.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'experiment' on GitHub by visiting:
remote:   https://github.com/kaverikadamm/project-alpha/pull/new/experiment
remote:
To https://github.com/kaverikadamm/project-alpha.git
 * [new branch]      experiment -> experiment
```



```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ echo "Updated Agenda" >> notes.txt

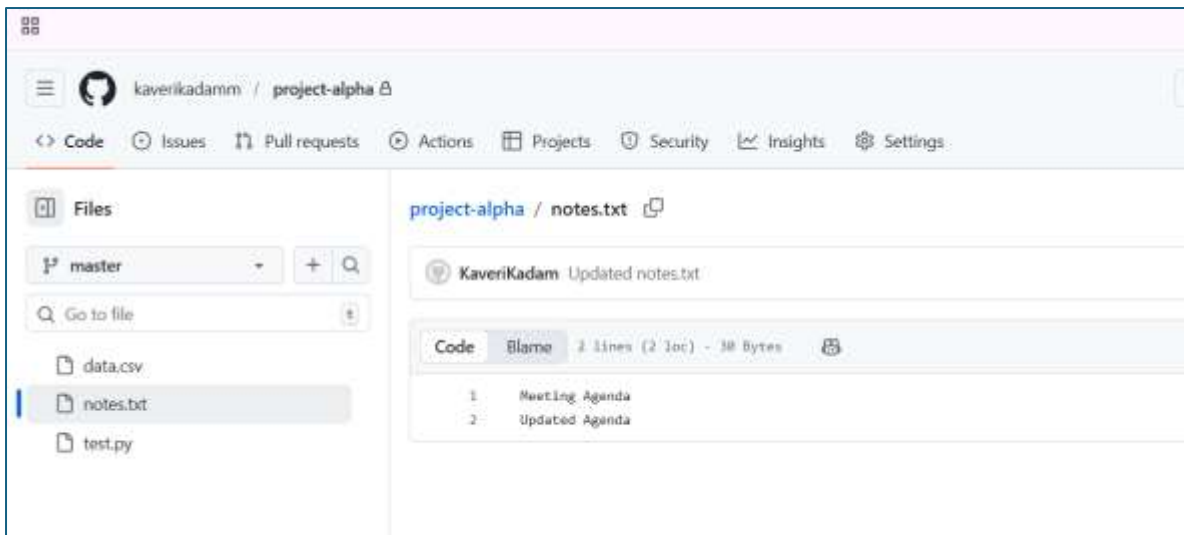
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git commit -m "Updated notes.txt"
[experiment fdd65fb] updated notes.txt
1 file changed, 1 insertion(+)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (experiment)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ git merge experiment
Updating eb8e476..fdd65fb
Fast-forward
 notes.txt | 1 +
 test.py   | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 test.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha (master)
$ git push origin master
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/project-alpha.git
 eb8e476..fdd65fb master -> master
```



7. Switch Context Mid-Task

Problem: You're halfway through a feature when asked to fix a critical bug in `main`. How do you save your work?

```
kaver@KAVERI MINGW64 ~
$ cd d:

kaver@KAVERI MINGW64 /d
$ cd MCA/"Sem II"/"DevOps Lab"

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q7

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q7

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q7/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ echo "Initial code" > app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git add app.js
warning: in the working copy of 'app.js', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git commit -m "Initial commit"
[master (root-commit) 87e718c] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git checkout -b feature
Switched to a new branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (feature)
$ echo "work in progress" >> app.js
```



```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (feature)
$ git stash
warning: in the working copy of 'app.js', LF will be replaced by C
RLF the next time Git touches it
Saved working directory and index state WIP on feature: 87e718c In
itial commit

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (feature)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ echo "Bug fixed" >> app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git add app.js
warning: in the working copy of 'app.js', LF will be replaced by C
RLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git commit -m "Fixed critical bug"
[master 77103ba] Fixed critical bug
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (master)
$ git checkout feature
Switched to branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q7 (feature)
$ git stash pop
On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working direc
tory)
        modified:   app.js

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (7f570f950ffb521557431a1776217a33d586d339)

```

8. Publish a Local Branch

Problem: Create a local Git repository with two branches (featureA and featureB). Each branch must contain two distinct files. Push both branches to a remote repository on GitHub.

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir myrepo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd myrepo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/myrep
o/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Initial file">>Reame.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add .
warning: in the working copy of 'Reame.md', LF will be replaced by
CRLF the next time Git touches it

```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "Initial commit"
[master (root-commit) ab4ce0b] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 Reame.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git checkout -b featureA
Switched to a new branch 'featureA'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureA)
$ echo "File A1" > fileA1.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureA)
$ echo "File A2" > fileA2.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureA)
$ git add .
warning: in the working copy of 'fileA1.txt', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of 'fileA2.txt', LF will be replaced
by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureA)
$ git commit -m "Added files in featureA"
[featureA 3e46474] Added files in featureA
2 files changed, 2 insertions(+)
create mode 100644 fileA1.txt
create mode 100644 fileA2.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureA)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git checkout -b featureB
Switched to a new branch 'featureB'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ echo "File B1" > fileB1.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ echo "File B2" > fileB2.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ git add .
warning: in the working copy of 'fileB1.txt', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of 'fileB2.txt', LF will be replaced
by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ git commit -m "Added files in featureB"
[featureB 9da98bd] Added files in featureB
2 files changed, 2 insertions(+)
create mode 100644 fileB1.txt
create mode 100644 fileB2.txt

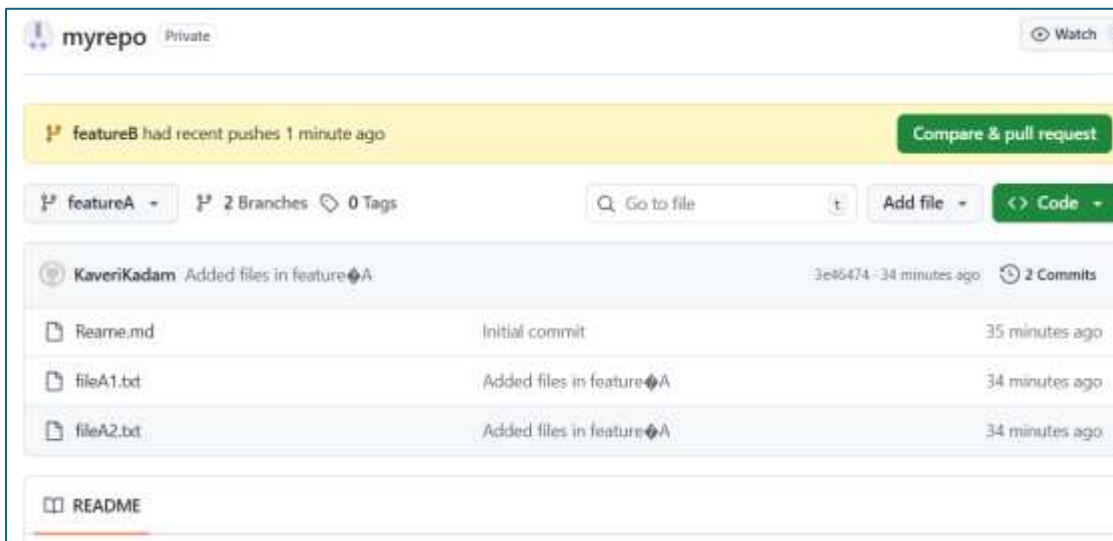
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ git remote add origin https://github.com/kaverikadamm/myrepo.git
```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ git push -u origin featureA
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (7/7), 533 bytes | 266.00 kiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/myrepo.git
 * [new branch]      featureA -> featureA
branch 'featureA' set up to track 'origin/featureA'.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (featureB)
$ git push -u origin featureB
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 333 bytes | 111.00 kiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'featureB' on GitHub by visiting
:
remote:      https://github.com/kaverikadamm/myrepo/pull/new/featureB
remote:
To https://github.com/kaverikadamm/myrepo.git
 * [new branch]      featureB -> featureB
branch 'featureB' set up to track 'origin/featureB'.

```



9. Rewrite Commit History

Problem: You made three messy commits for a feature. How do you combine them into one clean commit?

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q9

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q9

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q9/.git/

```



```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ echo "Line 1" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "commit 1"
[master (root-commit) 8bcc030] commit 1
1 file changed, 1 insertion(+)
create mode 100644 file.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ echo "Line 2" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "commit 2"
[master bb18de9] commit 2
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ echo "Line 3" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "commit 3"
[master 9a103d5] commit 3
1 file changed, 1 insertion(+)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git log --oneline
9a103d5 (HEAD -> master) commit 3
bb18de9 commit 2
8bcc030 commit 1
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git reset --soft HEAD~2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "Added complete feature"
[master 37a851a] Added complete feature
1 file changed, 2 insertions(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git log --oneline
37a851a (HEAD -> master) Added complete feature
8bcc030 commit 1
```


10. Roll Back to a Stable Version

Problem: The latest release (`v2.0`) has a critical bug. How do you revert to `v1.5`?

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ echo "Stable version" > app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git add app.txt
warning: in the working copy of 'app.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "Version 1.5 stable"
[master 450c0ed] Version 1.5 stable
1 file changed, 1 insertion(+)
create mode 100644 app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git tag v1.5

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ echo "Buggy code" >> app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git add app.txt
warning: in the working copy of 'app.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git commit -m "Version 2.0 with bug"
[master bba3031] Version 2.0 with bug
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git tag v2.0
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q9 (master)
$ git reset --hard v1.5
HEAD is now at 450c0ed Version 1.5 stable
```

11. Compare Branches

Problem: You need to see what's changed between `main` and `dev` before merging. How?

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd myrepo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Version 1" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "Initial commit"
[master 37a521a] Initial commit
2 files changed, 1 insertion(+), 3 deletions(-)
delete mode 100644 feature.txt
create mode 100644 file.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git checkout -b dev
Switched to a new branch 'dev'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (dev)
$ echo "Changes in dev branch" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (dev)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (dev)
$ git commit -m "Updated in dev"
[dev 8bdf38e] Updated in dev
1 file changed, 1 insertion(+)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (dev)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Changes in main branch" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "Updated in main"
[master b4d2192] Updated in main
1 file changed, 1 insertion(+)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git diff master dev
diff --git a/file.txt b/file.txt
index 537edda..553f217 100644
--- a/file.txt
+++ b/file.txt
@@ -1,2 +1,2 @@
 Version 1
-Changes in main branch
+Changes in dev branch
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git log master...dev
commit b4d219228f1060ae1ef905d237daf6e0feb3a7b0 (HEAD -> master)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 16:46:15 2026 +0530

    Updated in main

commit 8bdf38ea984af46af37dfe1bf861e4c6b941595f (dev)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 16:45:45 2026 +0530

    Updated in dev
```

12. Dealing with Commit

Problem: A developer made three commits but realizes the last commit contains incorrect changes. Perform the following:

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Line 1" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "commit 1"
[master 0c65d7b] commit 1
1 file changed, 1 insertion(+), 2 deletions(-)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Line 2" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "commit 2"
[master 54f2c91] commit 2
1 file changed, 1 insertion(+), 1 deletion(-)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ echo "Line 3" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by
CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git commit -m "commit 3"
[master 2eea28a] commit 3
1 file changed, 1 insertion(+), 1 deletion(-)
```

- View commit history

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git log --oneline
2eea28a (HEAD -> master) commit 3
54f2c91 commit 2
0c65d7b commit 1
```

- Undo the last commit but keep changes staged

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git reset --soft HEAD~1
```


- Undo another commit and move changes to working directory

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git reset --mixed HEAD~1
Unstaged changes after reset:
M      file.txt
```

- Completely remove the last commit and changes

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git reset --hard HEAD~1
HEAD is now at b4d2192 Updated in main
```

- How to restore a commit, if required.

Check lost commits:

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git reflog
b4d2192 (HEAD -> master) HEAD@{0}: reset: moving to HEAD~1
0c65d7b HEAD@{1}: reset: moving to HEAD~1
54f2c91 HEAD@{2}: reset: moving to HEAD~1
2eea28a HEAD@{3}: commit: commit 3
54f2c91 HEAD@{4}: commit: commit 2
0c65d7b HEAD@{5}: commit: commit 1
```

Restore:

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/myrepo (master)
$ git reset --hard 2eea28a
HEAD is now at 2eea28a commit 3
```

13. Write git commands to: create local repo project-alpha; add notes.txt/data.csv; commit; create branch experiment; add test.py with print("Hello"); commit.

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab (master)
$ mkdir project-alpha2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab (master)
$ cd project-alpha2

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (master)
$ echo "Meeting Agenda" > notes.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (master)
$ touch data.csv

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (master)
$ git add .
warning: in the working copy of 'project-alpha2/notes.txt', LF will be replaced by CRLF the next time Git touches it
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (master)
$ git checkout -b experiment
Switched to a new branch 'experiment'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (experiment)
$ echo "print('Hello')" > test.py

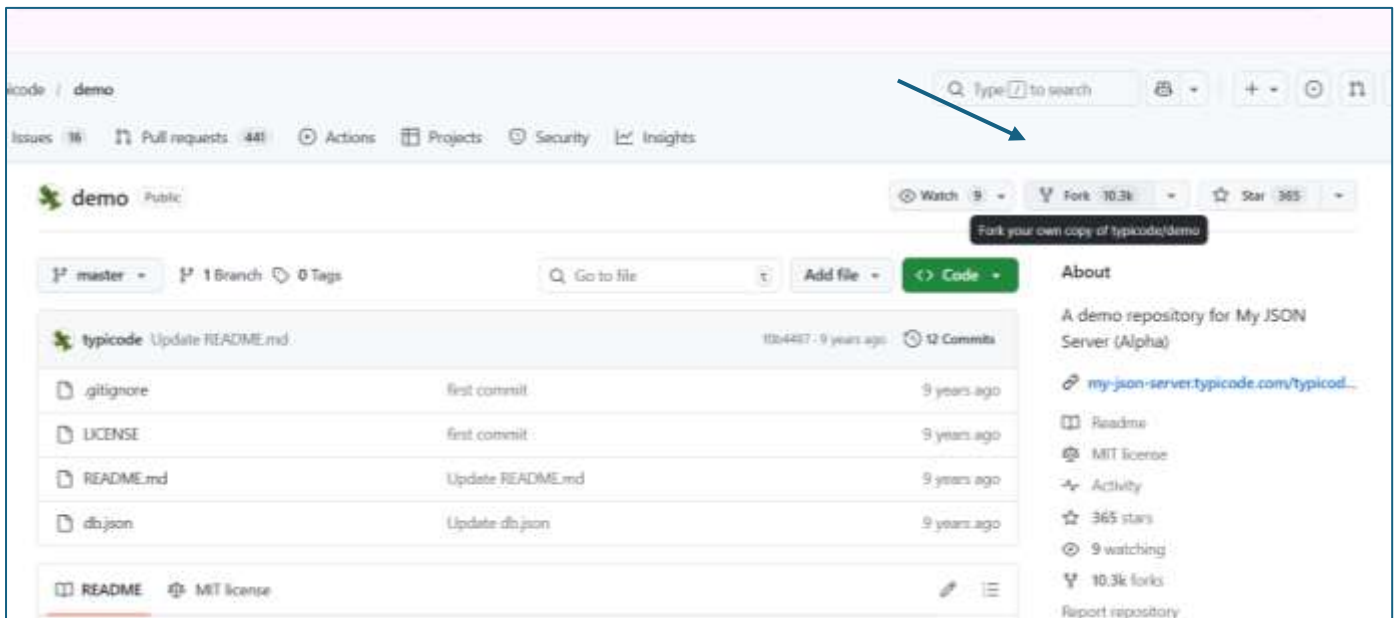
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (experiment)
$ git add test.py
warning: in the working copy of 'project-alpha2/test.py', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/project-alpha2 (experiment)
$ git commit -m "Added test.py in experiment branch"
[experiment (root-commit) d3c962f] Added test.py in experiment branch
3 files changed, 2 insertions(+)
create mode 100644 project-alpha2/data.csv
create mode 100644 project-alpha2/notes.txt
create mode 100644 project-alpha2/test.py
```

14. Collaborate via Fork

Problem: You want to contribute to an open-source project. How do you set up a fork? **Commands Needed:** Fork on GitHub, `git clone`, `git remote add upstream`.

Search any repository and click on fork.



Give any name (demorepo here) and click on fork

Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner *

Repository name *

kaverikadann

/ demorepo

demorepo is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

A demo repository for My JSON Server (Alpha)

44 / 350 characters

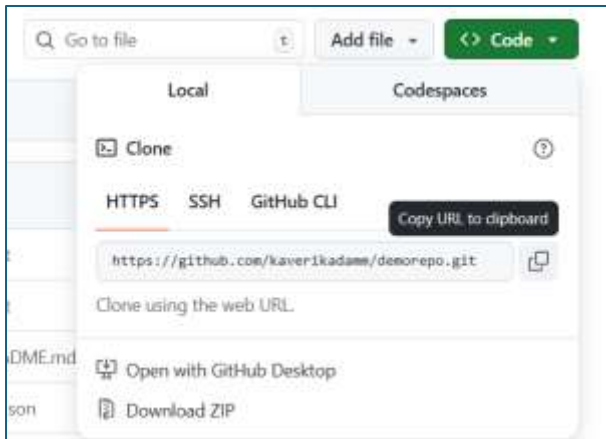
☒ Copy the `master` branch only

Contribute back to typicode/demo by adding your own branch. [Learn more.](#)

☐ You are creating a fork in your personal account.

Create fork

Go to code and copy the repository url



Clone your fork locally

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab (experiment)
$ mkdir Q14

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab (experiment)
$ cd Q14

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q14 (experiment)
$ git clone https://github.com/kaverikadamm/demorepo.git
Cloning into 'demorepo'...
remote: Enumerating objects: 39, done.
remote: Total 39 (delta 0), reused 0 (delta 0), pack-reused 39 (from 1
Receiving objects: 100% (39/39), 4.95 KiB | 506.00 KiB/s, done.
Resolving deltas: 100% (20/20), done.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q14 (experiment)
$ cd demorepo
```

Add original repo as upstream (copy url from original repo)

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q14/demorepo (master)
$ git remote add upstream https://github.com/typicode/demo.git
```

Verify Remotes

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q14/demorepo (master)
$ git remote -v
origin https://github.com/kaverikadamm/demorepo.git (fetch)
origin https://github.com/kaverikadamm/demorepo.git (push)
upstream https://github.com/typicode/demo.git (fetch)
upstream https://github.com/typicode/demo.git (push)
```

15. Rebase to Avoid Merge Commits

Problem: A developer is working on a feature branch while other developers update the master branch. Before pushing the feature, the developer wants to update the feature branch with the latest changes from master using rebase. Tasks to be performed -

- Create a repository
- Create a feature branch

- Add commits to both master and feature
- Rebase feature branch on master
- Perform rebase
- Resolve conflict manually
- Continue rebase

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q15

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q15

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q15/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ echo "Line 1" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git commit -m "Initial commit"
[master (root-commit) 25e137e] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git checkout -b feature
Switched to a new branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ echo "Feature work" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF
the next time Git touches it
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git commit -m "Feature commit"
[feature 3fef7a4] Feature commit
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ echo "Master update" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git commit -m "Master commit"
[master 18499b3] Master commit
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git checkout feature
Switched to branch 'feature'
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git rebase master
Auto-merging file.txt
CONFLICT (content): Merge conflict in file.txt
error: could not apply 3fef7a4... Feature commit
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue"
.
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git
rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict fa
lse"
could not apply 3fef7a4... # Feature commit
```

Conflict in file:

```
Line 1
<<<<<<< HEAD
Master update
=====
Feature work
>>>>>>> 3fef7a4 (Feature commit)
```

Remove lines:

```
Line 1
HEAD
Master update
Feature work
3fef7a4 (Feature commit)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git add file.txt
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git rebase --continue
```

(editor opens so :wq)

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (feature)
$ git log --oneline --graph
* dae9a6c (HEAD -> feature) Feature commit
* 18499b3 (master) Master commit
* 25e137e Initial commit
```

16. Track File History

Problem: You need to see how `README.md` evolved over time. How do you view its history?

Track file history

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ echo "Version 1" > README.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git commit -m "Initial README"
[master 1ba2b75] Initial README
1 file changed, 1 insertion(+)
create mode 100644 README.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ echo "Version 2" >> README.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF
the next time Git touches it
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git commit -m "Updated README"
[master d4da6b8] Updated README
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ echo "Version 3" >> README.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF
the next time Git touches it
```

View file history


```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git log README.md
commit d9ba9d530f84c526829052ac8fc8d22ec2f502cc (HEAD -> master)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 20:01:51 2026 +0530

    Final update

commit d4da6b8b662a67cfee0919ead3d5a079e7201d2f
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 19:42:34 2026 +0530

    Updated README

commit 1ba2b7597956da2e31a93900c003e38f08a728e3
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 19:42:07 2026 +0530

    Initial README

```

Changes between versions

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q15 (master)
$ git diff HEAD~1 README.md
diff --git a/README.md b/README.md
index 2ef8ab9..4322f9b 100644
--- a/README.md
+++ b/README.md
@@ -1,2 +1,3 @@
 Version 1
 Version 2
+Version 3

```

17. Revert a Public Commit

Problem: A commit pushed to `main` introduced a bug. How do you undo it without rewriting history?

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q17

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q17

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q17/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ echo "working code" > app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git add app.txt
warning: in the working copy of 'app.txt', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git commit -m "Initial commit"
[master (root-commit) ae47161] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 app.txt

```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ echo "Buggy code" >> app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git add app.txt
warning: in the working copy of 'app.txt', LF will be replaced by CRLF
the next time git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git commit -m "Bug Introduced"
[master 5d99868] Bug Introduced
1 file changed, 1 insertion(+)
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git remote add origin https://github.com/kaverikadamm/newrepo.git
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 453 bytes | 453.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/newrepo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git log --oneline
5d99868 (HEAD -> master, origin/master) Bug Introduced
ae47161 Initial commit
```

(below command opens editor)

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git revert HEAD
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q17 (master)
$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 282 bytes | 282.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/newrepo.git
 5d99868..be4f44e master -> master
```

18. Rename a Branch

Problem: The branch `bug/typo` should be renamed to `bug/readme-typo`. How?

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q18

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q18/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (master)
$ git checkout -b bug/typo
Switched to a new branch 'bug/typo'
```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/typo)
$ echo "Fixing typo in readme" >> readme.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/typo)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
    add

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/typo)
$ git add .
warning: in the working copy of 'readme.md', LF will be replaced by CRLF
the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/typo)
$ git commit -m "Fixed typo in readme"
[bug/typo (root-commit) cab3c73] Fixed typo in readme
1 file changed, 1 insertion(+)
create mode 100644 readme.md

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/typo)
$ git branch -m bug/typo bug/readme-typo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q18 (bug/readme-typo)
$ git branch
* bug/readme-typo

```

19. A developer is working on a feature branch and modifies app.js. Before completing the work, they must switch to the main branch to fix a bug. Perform the following:

- Save incomplete work using stash
- Switch to main branch
- Fix bug and commit changes
- Return to feature branch
- Restore stashed changes
- Checkout a Specific File

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q19

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q19

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q19/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ echo "Initial code" > app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git add app.js
warning: in the working copy of 'app.js', LF will be replaced by CRLF the
next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git commit -m "Initial commit"
[master (root-commit) 8b2b45d] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 app.js

```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git checkout -b feature
Switched to a new branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ echo "Feature work in progress" >> app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git stash
warning: in the working copy of 'app.js', LF will be replaced by CRLF t
he next time Git touches it
Saved working directory and index state WIP on feature: 8b2b45d Initial
commit

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git checkout master
Switched to branch 'master'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ echo "Bug fix" >> app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git add app.js
warning: in the working copy of 'app.js', LF will be replaced by CRLF t
he next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git commit -m "Bug fixed"
[master a2b1439] Bug fixed
1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (master)
$ git checkout feature
Switched to branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git stash pop
On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.js

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (81c0922a14603e4899e2c48295697a4c10bae561)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git checkout master -- app.js

```

20. Find a Deleted Line

Problem: A critical function was removed from `api.py`. How do you find it when it was deleted?


```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git checkout master -- app.js

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ echo "def important_function(): pass" > api.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git add api.py
warning: in the working copy of 'api.py', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git commit -m "Added important function"
[feature e9e27af] Added important function
 2 files changed, 2 insertions(+)
 create mode 100644 api.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ echo "# some other code" >> api.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git add api.py
warning: in the working copy of 'api.py', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git commit -m "Updated api"
[feature 1472299] Updated api
 1 file changed, 1 insertion(+)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ echo "# function removed" > api.py

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git add api.py
warning: in the working copy of 'api.py', LF will be replaced by CRLF the next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git commit -m "Removed Important Function"
[feature 3df2920] Removed Important Function
 1 file changed, 1 insertion(+), 2 deletions(-)
```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git log -p api.py
commit 3df292041d71e3fdf37696a7400bf9027c05576f (HEAD -> feature)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 21:55:41 2026 +0530

    Removed Important Function

diff --git a/api.py b/api.py
index 843de85..152b549 100644
--- a/api.py
+++ b/api.py
@@ -1,2 +1 @@
-def important_function(): pass
-# some other code
+# function removed

commit 1472299d15c407e03fb8aeda231a99849746dcf3
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 21:54:29 2026 +0530

    Updated api

diff --git a/api.py b/api.py
index 1790f47..843de85 100644
--- a/api.py
+++ b/api.py
@@ -1 +1,2 @@
 def important_function(): pass
+# some other code

commit e9e27af94395d393369c9481c9e09e16f9b18e7e
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 21:54:00 2026 +0530

    Added important function

diff --git a/api.py b/api.py

```

Directly search deletion

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q19 (feature)
$ git log -S "important_function" api.py
commit 3df292041d71e3fdf37696a7400bf9027c05576f (HEAD -> feature)
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 21:55:41 2026 +0530

    Removed Important Function

commit e9e27af94395d393369c9481c9e09e16f9b18e7e
Author: KaveriKadam <a250007@famt.ac.in>
Date: Thu Mar 26 21:54:00 2026 +0530

    Added important function

```

21. Clone a Single Branch

Problem: The repository is huge, but you only need the `dev` branch. How do you clone it?

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir demo-repo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd demo-repo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/demo-repo/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (master)
$ git checkout -b dev
Switched to a new branch 'dev'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ echo "Dev branch code" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the ne
xt time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git commit -m "Dev commit"
[dev (root-commit) e6260b6] Dev commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git remote add origin https://github.com/kaverikadamm/demo-repo.git

```

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git push -u origin dev
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 226 bytes | 226.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/demo-repo.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git clone --branch dev --single-branch https://github.com/kaverikadamm/demo-
repo.git
Cloning into 'demo-repo'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/demo-repo (dev)
$ git branch
* dev

```

22. Override Remote Changes

Problem: You need to force-push your local `main` branch to overwrite a corrupted remote.

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q22

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q22

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q22/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ echo "Correct code" > app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git add .
warning: in the working copy of 'app.txt', LF will be replaced by CRLF the ne
xt time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git commit -m "Clean version"
[master (root-commit) e70e9f0] Clean version
1 file changed, 1 insertion(+)
create mode 100644 app.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git remote add origin https://github.com/kaverikadamm/demo-repo.git

```

Forcepush to overwrite remote

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git push origin master --force
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 225 bytes | 225.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/kaverikadamm/demo-repo/pull/new/master
remote:
To https://github.com/kaverikadamm/demo-repo.git
 * [new branch]      master -> master

```

23. Sync with a Remote Repository

Problem: Your teammate pushed changes to GitHub, but your local repo is outdated. How do you update it?

Sync with Remote Repository

Pull latest changes

```

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git remote add origin https://github.com/kaverikadamm/demo-repo.git

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git pull origin master
From https://github.com/kaverikadamm/demo-repo
 * branch          master      -> FETCH_HEAD
 * [new branch]    master      -> origin/master
Already up to date.

```


24. Fix a Commit Message

Problem: The last commit message says “Fixed issuw”. How do you correct the typo?

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ echo "Some code" > file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the n
ext time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git commit -m "Fixed issuw"
[master 5255fc3] Fixed issuw
1 file changed, 1 insertion(+)
create mode 100644 file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q22 (master)
$ git commit --amend -m "Fixed issue"
[master 3d5900e] Fixed issue
Date: Thu Mar 26 22:14:13 2026 +0530
1 file changed, 1 insertion(+)
create mode 100644 file.txt
```

If already pushed to github:

git push origin master --force

25. Fork a GitHub repository, clone it locally, create a custom branch, apply modifications, commit, push, and create a pull request.

Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner *

 kaverikadamm ▾

Repository name *

 forkdemo is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

191 / 350 characters

☒ Copy the `main` branch only

Contribute back to open-telemetry/opentelemetry-demo by adding your own branch. [Learn more.](#)

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ mkdir Q25

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q25

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25
$ git clone https://github.com/kaverikadamm/forkdemo.git
Cloning into 'forkdemo'...
remote: Enumerating objects: 14568, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 14568 (delta 1), reused 0 (delta 0), pack-reused 14564 (from 2)
Receiving objects: 100% (14568/14568), 50.90 MiB | 2.37 MiB/s, done.
Resolving deltas: 100% (9232/9232), done.
```

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25
$ cd forkdemo

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25/forkdemo (main)
$ git remote add upstream https://github.com/open-telemetry/opentelemetry-demo.git

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25/forkdemo (main)
$ git checkout -b my-feature
Switched to a new branch 'my-feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25/forkdemo (my-feature)
$ echo "My contribution" >> file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25/forkdemo (my-feature)
$ git add .
git commit -m "Added my feature"
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[my-feature 30a6b7f] Added my feature
1 file changed, 1 insertion(+)
create mode 100644 file.txt

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q25/forkdemo (my-feature)
$ git push origin my-feature
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'my-feature' on GitHub by visiting:
remote:   https://github.com/kaverikadamm/forkdemo/pull/new/my-feature
remote:
To https://github.com/kaverikadamm/forkdemo.git
 * [new branch]      my-feature -> my-feature
```

26. Git Branching

Problem: You need to restore `index.html` from the `main` branch without switching branches.

```
kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab
$ cd Q26

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26
$ git init
Initialized empty Git repository in D:/MCA/SEM II/DevOps Lab/Q26/.git/

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ echo "<h1>Main Version</h1>" > index.html

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the
next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git commit -m "Main version"
[master (root-commit) 5104491] Main version
1 file changed, 1 insertion(+)
create mode 100644 index.html

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (master)
$ git checkout -b feature
Switched to a new branch 'feature'

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (feature)
$ echo "<h1>Feature Version</h1>" > index.html

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (feature)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the
next time Git touches it

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (feature)
$ git commit -m "Feature version"
[feature d0d72c8] Feature version
1 file changed, 1 insertion(+), 1 deletion(-)

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (feature)
$ git checkout master -- index.html

kaver@KAVERI MINGW64 /d/MCA/Sem II/DevOps Lab/Q26 (feature)
$ cat index.html
<h1>Main Version</h1>
```